DHBW Ravensburg

Intelligence Engineering

Solving Real World Problems At Scale

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LECTURE II

Introduction to Machine Learning Systems Design

This lecture in one slide

II Introduction to Machine Learning Systems Design

Business and ML Objectives

Requirements for ML Systems

Iterative Process

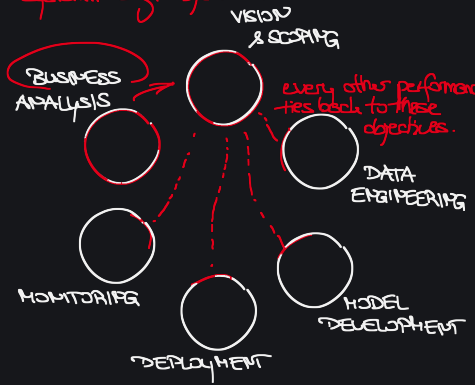
Framing ML Problems

Mind Versus Data

1

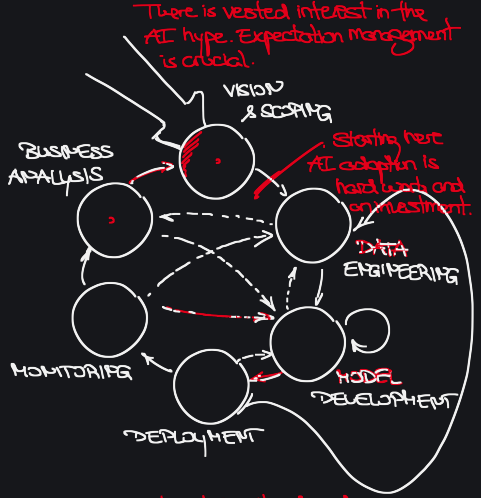
Business and ML Objectives

First Guidance on Developing the System through Objectives.



"The sole purpose of a business is to maximize profits for shareholders." - Milton Friedman

VISION & SCOPING
BUSINESS ANALYSIS
DATA ENGINEERING
MODEL DEVELOPMENT
DEPLOYMENT
MONITORING


There is vested interest in the AI hype. Expectation management is crucial.
VISION & SCOPING
BUSINESS ANALYSIS
Starting next AI adoption is hard work and an investment.
DATA ENGINEERING
MONITORING
MODEL DEVELOPMENT
DEPLOYMENT
Returns on investment heavily depend on maturity of adoption.
Development cycles and skills don't develop over-time.

2

Requirements for ML Systems

- **RELIABILITY**
  Correct function at desired level of performance in the face of adversity.
- **SCALABILITY**
  With respect to complexity, throughput, and other types of growth.
- **MAINTAINABILITY**
  Structure workload, infrastructure and tool landscape so that you can keep up.
- **ADAPTABILITY**
  Capacity to discover potentials for performance improvement and allowing updates without service interruptions.

3

Iterative Process

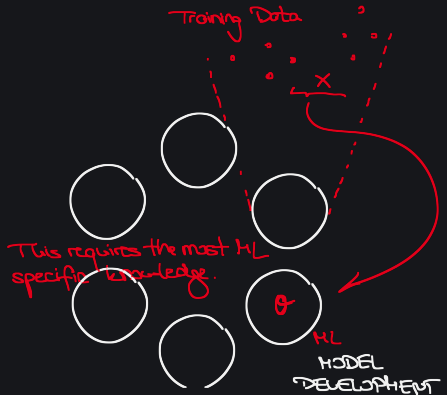
Developing ML systems is a cycle with a lot of back and forth!
#ITERATIVE.
VISION & SCOPING
BUSINESS ANALYSIS
DATA ENGINEERING
MODEL DEVELOPMENT
DEPLOYMENT
MONITORING

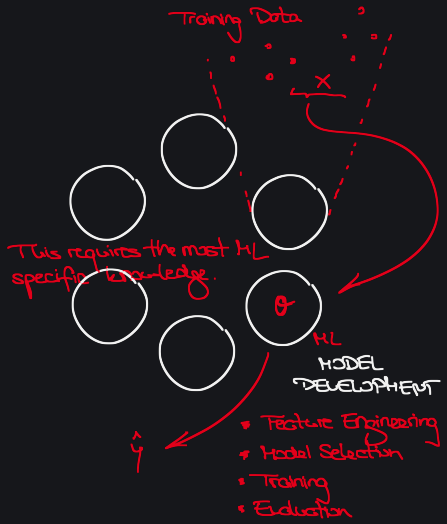
Objectives
Key Results
Constraints
VISION & SCOPING

Stakeholders: users, managers, humanity

We learn from data, so this is where we start with our craftmanship.
Different services, modalities, formats. - for examples
DATA ENGINEERING

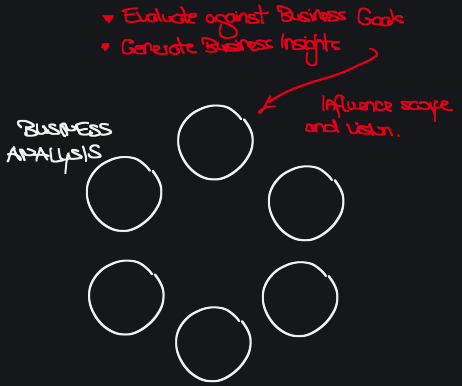
It is all about the labels


Training Data
X
This requires the most ML specific knowledge.
ML
MODEL DEVELOPMENT


Training Data
X
This requires the most ML specific knowledge.
ML MODEL DEVELOPMENT
• Feature Engineering
• Model Selection
• Training
• Evaluation
ŷ

You will never reach a point where your system is done.
At some point you need to expose your system.
DEPLOYMENT

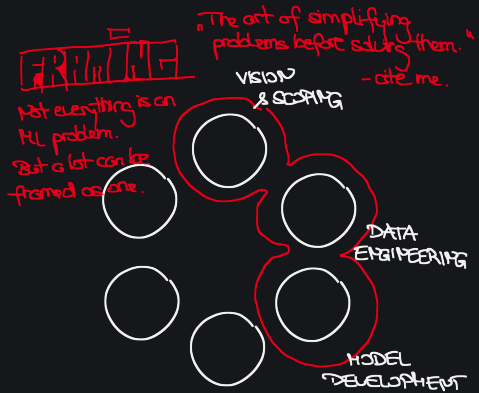
Changing requirements
Performance Decay
MONITORING
Capability to
stay adaptive
Changing environments



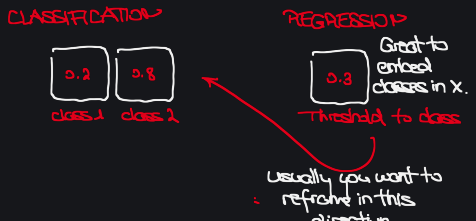
Business analysis closes the loop

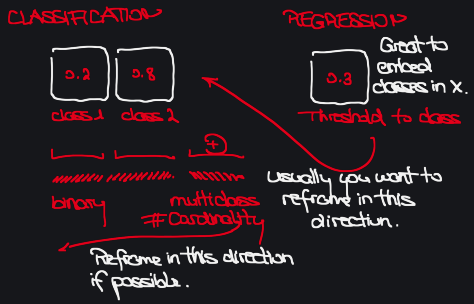
4

Framing ML Problems



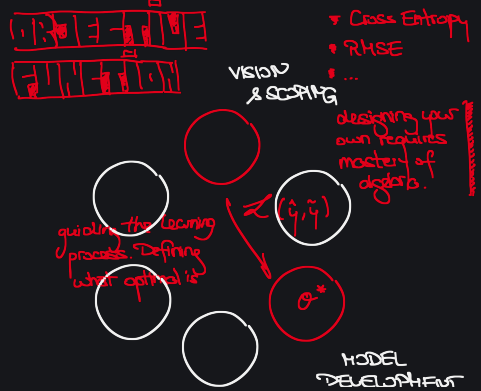
A lot can be framed as an ML problem


CLASSIFICATION
0.2
0.8
class 1
class 2
REGRESSION
0.3
Great to embed classes in X.
Threshold to class
Usually you want to reframe in this direction.


CLASSIFICATION
REGRESSION
0.2
0.8
0.3
Great to enforce classes in X.
class 1
class 2
Threshold to class
+
Usually you want to reframe in this direction.
binary
multiclass
#Cardinality
Reframe in this direction if possible.

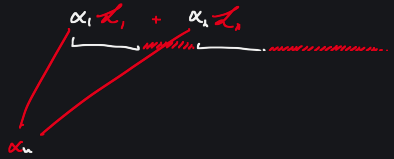
- Restrain from „buying" this with an increasing number of models.
  Hierarchical & Sequential systems are sophisticated but hard to maintain.

→ Talk is cheap. Go and try those things on MNIST to sharpen your own intuition.

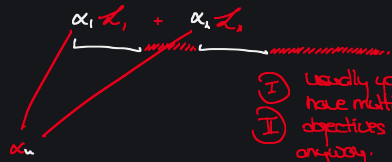


The objective function defines what optimal is

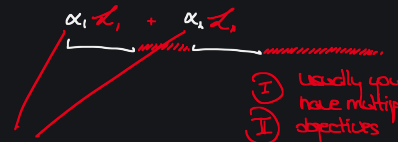
THE EFFECTIVENESS OF DECOUPLING OBJECTIVES.



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THE EFFECTIVENESS OF DECOUPLING OBJECTIVES.



I) Usually you have multiple objectives anyway.

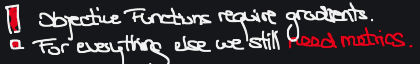
II) This is a clean way to state them explicitly.

#Pareto Optimization

Trivial to weight objectives while training.

Everytime you touch is a retrain.

Beautiful opportunity to train two models and adaptively weight during inference.


Objective Functions require gradients.
For everything else we still need metrics.

5

Mind Versus Data


As data flows through models. During the inference its key
Intelligent architectural designs frozen within the current paradigm. Data-centric.
MODEL success heavily depends on data.
Biggest leverage in managing and improving data quality and quantity.
DATA utilization again is tied to compute.

"We don't have better algorithms. We just have better data." - Norvig (Director Search @Google)



Remember how more data is a regularization method, driving generalization of your models?

References